

Model Paper - Database Management Systems

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Q1

Consider the following Enhanced Entity-Relationship (EER) diagram:

Note: The diagram uses (min, max) cardinality notation where:

- (1,1) indicates one-to-one relationship (each entity participates exactly once)
- (1,N) or (1,*) indicates one-to-many relationship (one entity can relate to many)
- (0,1) indicates optional participation (zero or one)
- (0,N) or (0,*) indicates optional many participation (zero or many)

A university is developing a database system to manage its academic programs. The main entities include 'Department', 'Professor', and 'Course'. Each department is characterized by a unique DepartmentID and DepartmentName. Professors are identified by ProfessorID and ProfessorName, and each course has a CourseID, CourseName, and can have multiple prerequisites. A department offers many courses, and each course is taught by one or more professors; additionally, professors can belong to multiple departments. Furthermore, some courses are specialized for graduate levels (GraduateCourse). Each relationship must account for applicable constraints such as cardinalities and participation

Figure: EER



- a) Convert the following EER model into the relational model. Indicate the primary keys and the foreign keys of the resulting relations clearly.
- b) What is the best option for mapping the ISA hierarchies in the diagram? Justify your answer.

Q2

Consider a relation $R(A, B, C, D, E)$ with the following set of functional dependencies F over R : $F = \{AB \rightarrow C, B \rightarrow D, C \rightarrow E, A \rightarrow D, DE \rightarrow B\}$. This relation represents a product inventory system where attributes A and B represent product identifiers, C denotes product name, D shows the location in the warehouse, and E indicates the supplier of the product. Analyze the given relation in terms of its keys and normalization requirements based on the outlined functional dependencies.

- a) Find all the keys in relation R using the attribute closure method.
- b) Is R in 3NF? Give reasons for your conclusion.
- c) Is R in BCNF? Give reasons for your conclusion. If R is not in BCNF, convert it to a set of BCNF relations.

Q3

A university is developing a database system to manage its course offerings, student enrollments, and faculty assignments. The database design includes the following relations: `Courses(course_id: integer, course_name: string, credits: integer)`, `Students(student_id: integer, student_name: string, major: string)`, and `Enrollments(student_id: integer, course_id: integer, semester: string, grade: string)`. Each course can have multiple students enrolled, and a student can enroll in multiple courses. Moreover, a faculty member can teach multiple courses, and each course is assigned to one faculty member. The design must also include constraints such as primary keys, foreign keys, and any necessary rules to maintain integrity.

- a) In Type 2 Driver, JDBC API calls are converted to native Java API calls. Accept or refute the above statement justifying your answer.
- b)

```
String sql = "SELECT * FROM employees WHERE department = ?";
PreparedStatement pstmt = connection.prepareStatement(sql);
pstmt.setString(1, "IT");
ResultSet rs = pstmt.executeQuery();
```

There are several different statements in the JDBC API to retrieve result sets based on different requirements. Which type of statements is used in the code segment shown above? Briefly explain when this type of statement will be used.

- c) Briefly explain the options available for transferring data between two sources such as between tables and servers.
- d) Briefly explain the importance of foreign keys in the context of the `Enrollments` table and how they contribute to data integrity.
- e) **A university is establishing role-based access control for its database system that manages Courses, Students, and Enrollments. Sarah is the senior database administrator responsible for overall database**

administration. Emily is a junior database administrator responsible for managing the Enrollments database. Nathan is a database developer responsible for creating and maintaining database objects. Michael is a data entry operator responsible for entering and updating student enrollment data.

- i. Write a T-SQL statement to create a login to Sarah with windows authentication.
- ii. Provide Sarah with the necessary permissions to handle all administrative tasks within the server using a fixed server role.
- iii. Assuming Emily's username is 'emily.e', write T-SQL statements to assign her the responsibility of managing the Enrollments database using a user-defined server role.
- iv. Assuming Nathan's login name is 'nathan.n', write T-SQL statements to grant him permission to create tables and databases.
- v. Assuming Michael's login name is 'michael.m', write T-SQL statements to allow him to perform data entry tasks on the Students and Enrollments databases.

Q4

Consider the following schema of a database designed for a Hospital: Patient (patientId: int, name: varchar(100), address: varchar(150), dob: date, phone: varchar(15)) Doctor (doctorId: int, name: varchar(100), specialty: varchar(50), phone: varchar(15)) Appointment (appointmentId: int, patientId: int, doctorId: int, appointmentDate: date, status: varchar(20)) Fee (feeId: int, appointmentId: int, amount: real, paymentStatus: varchar(20)) The 'Patient' table stores personal information about patients, the 'Doctor' table keeps records of doctors and their specialties, the 'Appointment' table contains details about scheduled appointments, and the 'Fee' table logs the fees corresponding to each appointment along with their payment status. The 'Doctor' table holds details about doctors, including unique doctor ID, name, specialization, and contact information. The 'Fee' table contains relevant information for the database system.

a) Write SQL Queries to perform the following:

- i. Find the name and phone number of the doctor who has an appointment scheduled on '2023-10-01'.
 - ii. Find the patient who has the highest total amount compared to all other patients. Find the patientId and name.
 - iii. Find the name and name from Doctor, Patient, and Appointment where Appointment.appointmentDate is not null.
- b) Create a function to calculate the total fee amount for a given patient. This function should account for fee with a late penalty of 10% applied to those marked as 'overdue' payment status.
- c) Create a trigger that automatically updates the 'amount' column in the 'Fee' table whenever a new fee record is added, or an existing fee record is updated. The 'amount' column should contain the total fee amount for each patient. Ensure the trigger adjusts the 'amount' column accurately to reflect any partial payments made by patient. The patient pays only 50% of the total fee amount as an initial payment when the payment status is marked as 'Partial'.